Exercise 14

Applications on definite Integral

From the school book | Understand | Apply | Higher Order Thinking Skills

First Areas in the plane

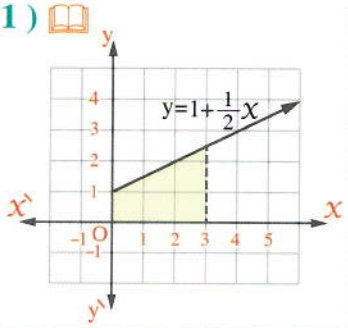
1 **Find the area of the region under the curve of the function y and above the X-axis in the given interval in each of the following :**

(1) $y = 2X + 3$ above the interval $[-1, 4]$ (2) $y = |X - 3|$ above the interval $[2, 5]$

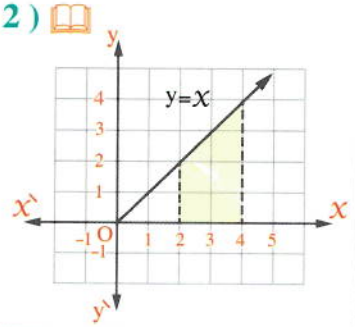
(3) $y = 3X^2$ above the interval $[-3, 1]$

2 **Write down the definite integration which gives the shaded area in each of the following and calculate its value :**

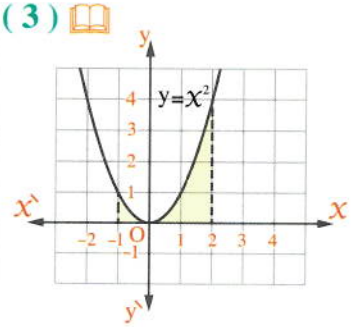
(1)



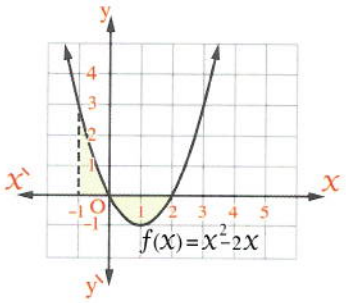
(2)



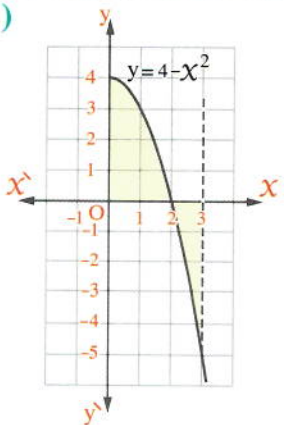
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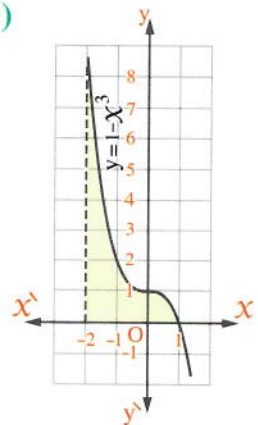
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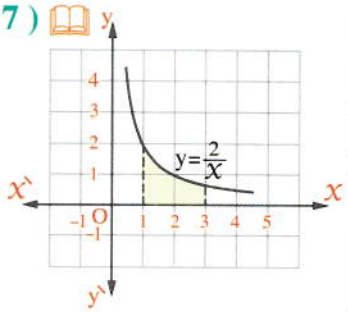
(5)



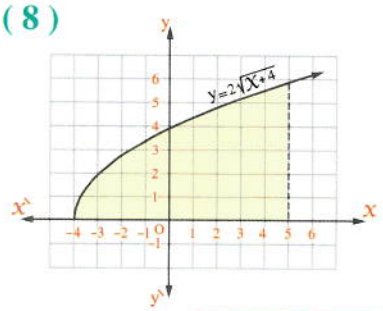
(6)



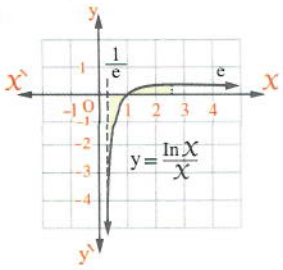
(7)



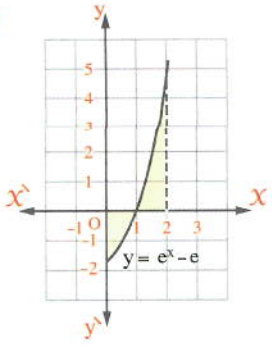
(8)



(9)



(10)



3 **Find in each of the following the area of the plane region bounded by :**

(1) The straight lines : $x + 2y = 9$, $x = 1$, $x = 3$, $y = 0$ « 7 square units »

(2) The curve : $y = x^2 - 5x$ and the x-axis « $20\frac{5}{6}$ square units »

(3) The curve : $y = 5 - x^2$ and the x-axis and the two straight lines : $x = -2$ and $x = 1$ « 12 square units »

(4) The curve : $y = 3 - 2x - x^2$ and x-axis « $\frac{32}{3}$ square units »

(5) The curve : $y = 3 + 2x - x^2$ and the straight lines $x = -1$, $x = 4$, $y = 0$ « 13 square units »

(6) The curve of the function f : $f(x) = 3x^2 + 1$ and the x-axis and the two straight lines : $x = -1$, $x = 2$ « 12 square units »

(7) The curve : $y = \sqrt{4x}$ and the straight line $x = 9$ and $y = 0$ « 36 square units »

(8) The curve : $y = \sqrt{x + 4}$ and the straight lines : $x = 0$, $x = 5$ and $y = 0$ « $\frac{38}{3}$ square units »

(9) The curve : $y = \sqrt{1 - x}$ and the straight line : $x = -3$ and $y = 0$ « $5\frac{1}{3}$ square units »

(10) The curve f : $f(x) = \sqrt[3]{2x + 2}$ and the straight line : $x = 3$ and above the x-axis. « 6 square units »

(11) The curve : $y = \frac{4}{x^2}$ and the straight lines $x = 1$, $x = 4$, $y = 0$ « 3 square units »

(12) The curve : $y = e^x$ and the straight lines : $x = 0$, $x = \ln 4$ and $y = 0$ « 3 square units »

(13) The curve : $y = \sin x$ and the straight lines : $x = 0$, $x = \frac{\pi}{2}$ and $y = 0$ « 1 square unit »

4 **Using the area under the curve , find the value of each of the following :**

(1) $\int_0^2 \sqrt{4-x^2}\,dx$ « π square units »

(2) $\int_{-3}^3 \sqrt{9-x^2}\,dx$ « $\frac{9}{2}\pi$ square units »

(3) $\int_{-7}^0 \sqrt{49-x^2}\,dx$ « $\frac{49}{4}\pi$ square units »

(4) $\int_0^1 \sqrt{4-x^2}\,dx$ « $\frac{\sqrt{3}}{2}+\frac{\pi}{3}$ square units »

5 **Choose the correct answer from the given ones :**

(1) *(2nd session 2019)* The area of the region bounded by the straight lines $y = x$, $x = 2$, $y = 0$ equals ..... square units.

(a) $\frac{1}{2}$ (b) 1 (c) 2 (d) 4

(2) *(Speciman 2025)* The area of the region bounded by the straight lines : $y = \frac{1}{2}x + 4$, $x = -1$, $x = 3$ and the x-axis equals ..... square units.

(a) 15 (b) 16 (c) 17 (d) 18

(3) *(Speciman 2025)* The area of the region bounded by the curve $y = x^2 + 3$ and the two straight lines $x = -2$, $x = 5$ and x-axis equals ..... square units.

(a) $\frac{196}{3}$ (b) 154 (c) $\frac{154}{3}$ (d) 48

(4) *(2nd session 2019)* The area of the region bounded by the curve $y = x^3$ and the straight lines $y = 0$, $x = 2$ equals ..... square unit.

(a) 8 (b) 4 (c) 2 (d) 1

(5) $\int_{-1}^1 \sqrt{16-16x^2}\,dx =$ ..... square unit.

(a) 16π (b) π (c) 2π (d) 4π

(6) *(2nd Session 2024)* The area of the region bounded by the curve $y = \sqrt{x}$ and the two straight lines $y = 0$, $x = 4$ equals ..... square unit.

(a) $\frac{16}{3}$ (b) $\frac{8}{3}$ (c) $\frac{4}{3}\sqrt{2}$ (d) $\frac{8}{3}\sqrt{2}$

(7) The area of the region bounded by the curve $y = \sqrt{4-x^2}$ and x-axis equals ..... square units.

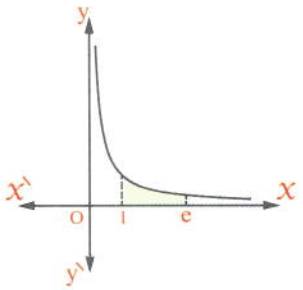
(a) 2 (b) 4 (c) 2π (d) 4π

(8) If f is a continuous function on the interval $[a, b]$ and A is the area bounded by the curve of the function $y = f(x)$, the x-axis and the two straight lines $x = a$, $x = b$, then A = ...

(a) $\int_a^b |y| \, dx$ (b) $\left|\int_a^b y \, dx\right|$ (c) $\int_a^b y \, dx$ (d) $\int_a^b |x| \, dy$

(9) In the opposite figure :

If $f(x) = \frac{1}{x}$, then the area of the shaded region = ... square units.

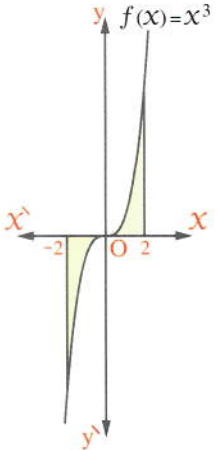


(a) 1 (b) e

(c) e^2 (d) 2

(10) *(1st Session 2024)* In the opposite figure :

The area of the shaded region = ... square units.



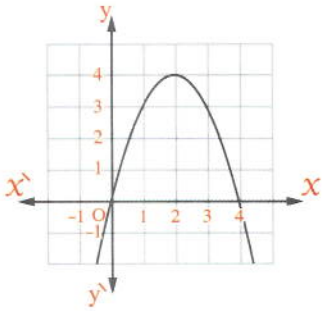
(a) 8

(b) 4

(c) 2

(d) zero

(11) The opposite figure represents the curve of the function $f : f(x) = -x(x-4)$, then all the following is true except .....



(a) $\int_0^1 f(x) \, dx = \int_3^4 f(x) \, dx$

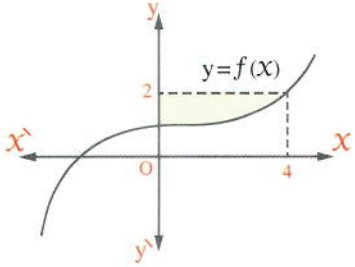
(b) $\int_0^4 f(x) \, dx = 2\int_0^2 f(x) \, dx$

(c) $\int_{-1}^0 |f(x)| \, dx = \int_0^1 f(x) \, dx$

(d) $\int_{-1}^0 f(x) \, dx = \int_4^5 f(x) \, dx$

(12) In the opposite figure :

If the area of the shaded region = 3 square units, then $\int_0^4 f(x) \, dx$ = ...

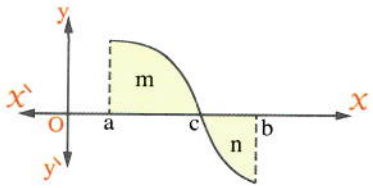


(a) 3 (b) 4

(c) 5 (d) 6

(13) In the opposite figure :

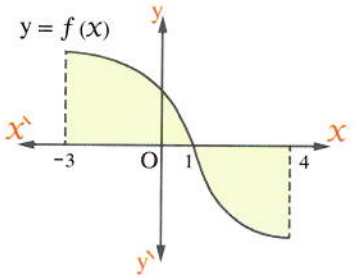
It is a part of a curve $f(x)$ in the interval $[a, b]$, then if the area m equals 5 square unit and area n equals 3 square unit , then $\int_a^b f(x)\,dx =$ .....



(a) -5 (b) 4 (c) 2 (d) 8

(14) In the opposite figure :

If $\int_{-3}^{4} f(x)\,dx = 12$ and the area of the shaded part = 28 square unit , then $\int_{1}^{4} f(x)\,dx =$ .....

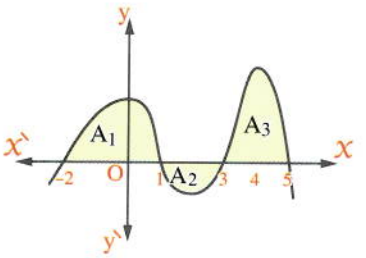


(a) 16 (b) -8

(c) 8 (d) 20

(15) In the opposite figure :

If $A_1 = 5$ square units , $A_2 = 2$ square units , $A_3 = 8$ square units , then : $\int_{-2}^{5} f(x)\,dx + \int_{-2}^{5} |f(x)|\,dx =$ .....

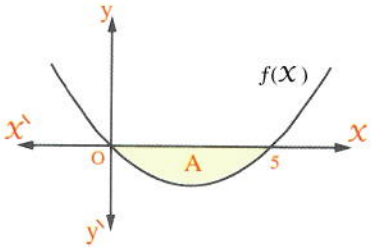


(a) 15 (b) 20

(c) 22 (d) 26

(16) In the opposite figure :

If the region A between the curve $f(x)$ and the x-axis equals 8 square units , then $\int_0^5 (1 - f(x))\,dx =$ .....

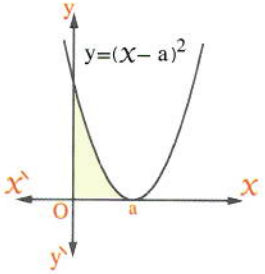


(a) 12 (b) 13

(c) 14 (d) 15

(17) In the opposite figure :

If the area of the shaded region = $\frac{8}{3}$ square units , then a = ...



(a) $\frac{1}{2}$ (b) 1

(c) $\frac{3}{2}$ (d) 2

(18) The area of the region bounded by the curve $x\,y = 4$, x-axis and the two straight lines $x = 1$, $x = 3$ equals .....

(a) 2 ln 3 (b) 4 ln 3 (c) 3 ln 3 (d) 3 ln 4

(19) *(Speciman 2025)* The area of the region bounded by the curve $y = e^x$ and y-axis , and the straight line $x = 2$ and above x-axis equals ..... unit area.

(a) e^2 (b) 2 e (c) $e^2 - 1$ (d) $e^2 - e$

(20) *(2nd Session 2021)* The area of the region bounded by the curve $y - 1 = e^x$ and x-axis and the straight lines $x = 0$, $x = k$ where $k > 0$ equals ..... square unit

(a) $e^k - k + 1$ (b) $e^k + k + 1$ (c) $e^k + k - 1$ (d) $e^k - k - 1$

(21) *(2nd Session 2022)* The area of the region bounded by the curve $y = f(x)$ whose parametric equations are $y = \sin\theta$, $x = \csc\theta$, x-axis and the two straight lines $x = 1$, $x = e$ equals ..... square unit.

(a) e (b) 1 (c) e^2 (d) 2

(22) *(2nd Session 2022)* The area of the region which is bounded by the curve $y - 3 = e^x$, x-axis and the two straight lines $x = 0$, $x = 1$ equals ..... square units.

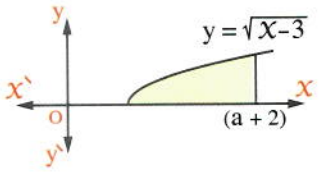
(a) e + 2 (b) e^3 (c) e + 3 (d) 2 e

(23) *(2nd Session 2021)* The area bounded by the curve $y = f(x)$ where $y = n - \frac{n^2}{4}$, $x = \frac{n}{2}$ and x-axis equals ..... square unit.

(a) $\frac{7}{3}$ (b) $\frac{3}{4}$ (c) $\frac{4}{3}$ (d) $\frac{3}{7}$

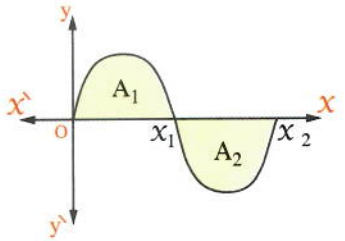
(24) *(1st Session 2022)* **In the opposite figure :**

If the area of the shaded part = $\frac{16}{3}$ square units , then a = ...



(a) 2 (b) 3 (c) 4 (d) 5

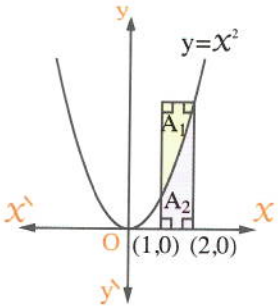
(25) The opposite figure represents the curve of the function f , then the area included between the curve of the function f and x-axis equals .....



(a) $\int_0^{x_2} f(x)\,dx$ (b) $\int_0^{x_1} f(x)\,dx + \int_{x_1}^{x_2} f(x)\,dx$

(c) $\int_0^{x_1} f(x)\,dx - \int_{x_1}^{x_2} f(x)\,dx$ (d) $\left|\int_0^{x_2} f(x)\,dx\right|$

(26) The opposite figure represents the curve of the function $f : f(x) = x^2$, then $\frac{A_1}{A_2}$ = ...



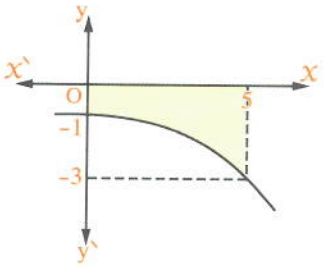
(a) $\frac{2}{7}$ (b) $\frac{3}{7}$

(c) $\frac{4}{7}$ (d) $\frac{5}{7}$

(27) In the opposite figure :

If the area of the shaded region = 10 square units.

, then $\int_0^5 [f'(x) + f(x)]\, dx$ = ...

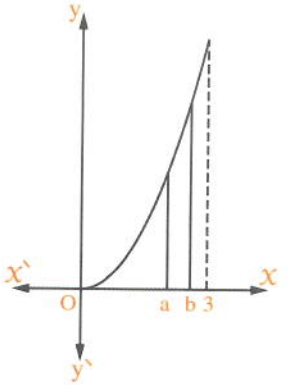


(a) – 4 (b) – 18

(c) – 12 (d) – 8

(28) In the opposite figure :

If the area under the curve $y = x^2$ and including between the two straight lines $x = 0$, $x = 3$ is divided into three equal parts by two straight lines $x = a$, $x = b$ then $a \times b$ = ...



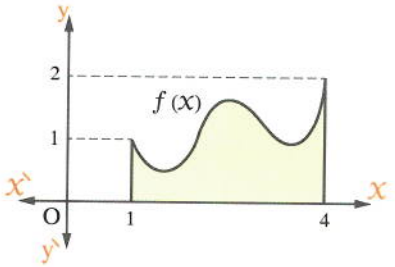
(a) $5\sqrt{2}$ (b) $6\sqrt{3}$

(c) $3\sqrt[3]{6}$ (d) $2\sqrt[3]{9}$

(29) ***(Speciman 2025)*** **In the opposite figure :**

If the area of the shaded region = 5 square units

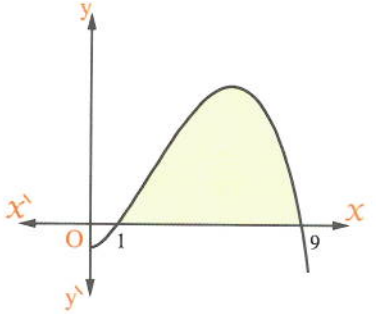
, then $\int_1^4 x\, f'(x)\, dx$ = ...



(a) 2 (b) 3

(c) 4 (d) 5

(30) ***(Speciman 2025)*** The opposite figure represents the curve of the function $y = f(x)$, and the area of the shaded region = 36 square unit



, then $\int_0^2 x^2 f(x^3 + 1)\, dx$ = ...

(a) 12 (b) 18

(c) 72 (d) 108

6 Find in square units the area of the region bounded by the curve of : $f(x) = (x-2)^3$ and the x-axis in the interval $[2, 4]$ « 4 square units »

7 An architect has designed an arc -like entryway of a hotel whose equation $y = -\frac{1}{2}(x-1)(x-7)$ where x in metres. How much does the glass cost if this entryway is covered by the glass which costs L.E. 1500 per square metre ? « L.E. 27000 »

8 If the cost of a squared metre of granite to cover the floor of a hotel corridors is L.E. 400 and five corridors have been already covered with granite and the area of each is bounded by the curve of the function f and the two straight lines $x = 0$ and $y = 0$ where $f(x) = 12 - \frac{1}{3}x^2$ Find the cost covering the five corridors. « L.E. 96000 »

9 Find the area of the region above x-axis bounded by the curve of the function $f : f(x) = \frac{4x}{x^2+1}$ and the straight line $x = 4$ « 2 ln (17) square units »

10 Find the area of the region bounded by the curve : $y = x^3 - 6x^2 + 8x$ and the x-axis. « 8 square units »

11 If $f :]-\infty, 3] \longrightarrow \mathbb{R}$ where $f(x) = x^3 - 4x$, find the area of the region above x-axis bounded by the curve of the function and x-axis. « $10\frac{1}{4}$ square units »

12 If $f(x) = x^3 - 3x + 3$, **find :**

(1) The absolute extrema value of the function f in the interval $f[0, 2]$ « 1 , 5 »

(2) The area of the region bounded by the curve of the function f and the straight lines $x = 0$, $x = 2$, $y = 0$ « 4 square unit »

13 Find the area bounded by the curve of the function $f : f(x) = (3-x)(x-1)^2$ and the two coordinate axes where $f(x) \geq 0$ « $\frac{9}{4}$ square units »

14 Find the area bounded by the curve of the function $f : f(x) = (x-1)(x-2)(x-3)$ and the two straight lines $x = 4$, $y = 0$ where $f(x) \geq 0$ « $2\frac{1}{2}$ square units »

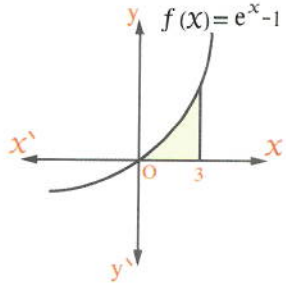
15 Find the area of the region bounded by the curve : $\sqrt{x} + \sqrt{y} = 1$ and the two straight lines $x = 0$, $y = 0$ « $\frac{1}{6}$ square units »

16 Use the definite integration to prove that the area of the triangle whose base length equals a and its height equals b is $\frac{1}{2}$ a b

17 Find the area bounded by the curve : $y = x(x-1)^2$ and the x-axis « $\frac{1}{12}$ square units »

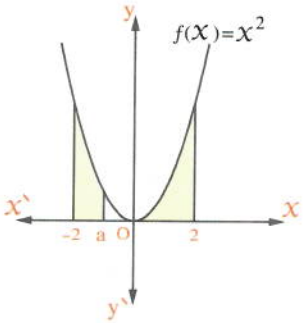
18 ***(1st Session 2023)*** **In the opposite figure :**

If the area of the shaded region equals $(a^3 - 4)$ square units , find the value of a. « e »



19 ***(2nd Session 2023)*** **In the opposite figure :**

If the area of the shaded region equals $\frac{15}{3}$ square units , find the value of a « – 1 »



Second volumes of revolutions solids

20 **Find the volume of the solid generated by revolving the region bounded by the given curves and straight lines a complete revolution about X-axis for each of the following :**

(1) $y = 2X + 1$, $y = 0$, $X = 0$, $X = 4$ « $121\frac{1}{3}\pi$ cubic unit »

(2) 📖 $y = 3 - X$, $X = 0$, $y = 0$ « 9π cubic unit »

(3) 📖 $f(X) = X^2 + 1$, $y = 0$, $X = -1$, $X = 1$ « $\frac{56}{15}\pi$ cubic unit »

(4) $y = \sqrt{X}$, $y = 0$, $X = 1$, $X = 4$ « $7\frac{1}{2}\pi$ cubic unit »

(5) $y = \sqrt{X+5}$, $y = 0$, $X = 1$, $X = 3$ « 14π cubic unit »

(6) $y = X^3 + 1$, $y = 0$, $X = 0$, $X = 1$ « $\frac{23}{14}\pi$ cubic unit »

(7) 📖 $y = \frac{1}{X}$, $X = 1$, $X = 4$, $y = 0$ « $\frac{3}{4}\pi$ cubic unit »

(8) $y = e^X$, $y = 0$, $X = 0$, $X = 2$ « $\frac{\pi}{2}(e^4 - 1)$ cubic unit »

(9) 📖 $y = |X|$, $X = -2$, $X = 4$, $y = 0$ « 24π cubic unit »

(10) $y = \sin 2X$, $y = 0$, $X = 0$, $X = \frac{\pi}{2}$ « $\frac{1}{4}\pi^2$ cubic unit »

(11) $f(X) = \sec X$, $y = 0$, $X = 0$, $X = \frac{\pi}{4}$ « π cubic unit »

(12) $f(X) = \csc X$, $y = 0$, $X = \frac{\pi}{4}$, $X = \frac{\pi}{2}$ « π cubic unit »

21 📖 Find the volume of the solid generated by revolving the plane region bounded by the curve of the function f and the X-axis and the two straight lines $X = 0$, $X = 3$ a complete revolution about X-axis given that $f(X) = X$ what is the name of the generated solid ? Show how you can check your answer geometrically. « 9π cubic unit »

22 Find the volume of the solid generated by revolving the region bounded by the curve $y = x^2$ and the x-axis and the straight line $x = 2$ a complete revolution about x-axis.

« $\frac{32}{5}\pi$ cubic unit »

23 Find the volume of the solid generated by revolving the region bounded by the curve $y = \sqrt{4 - x}$ and the two axes of coordinates a complete revolution about x-axis.

« 8π cubic unit »

24 Find the volume of the solid generated by revolving the region bounded by the curve $y = 4 - x^2$ and the two positive parts of the axes of coordinates a complete revolution about x-axis.

« $\frac{256}{15}\pi$ cubic unit »

25 Find the volume of the solid generated by revolving the region bounded by the curve $y^2 = 8x$ and the straight line $x = 2$ a half revolution about x-axis.

« 16π cubic unit »

26 Find the volume of the solid generated by revolving the region bounded by the curve $y^2 = 4 - x$ and the two positive parts of the axes of coordinates a complete revolution about x-axis.

« 8π cubic unit »

27 Find the volume of the solid generated by revolving the region bounded by the x-axis and the given curve a complete revolution about x-axis in each of the following :

(1) $f(x) = 1 - x^2$

(2) 📖 $f(x) = 2x - x^2$

(3) $f(x) = x^3 - x$

« $\frac{16}{15}\pi$, $\frac{16}{15}\pi$, $\frac{16}{105}\pi$ cubic unit »

28 If a is the region bounded by the curve $x\,y = 4 + x^2$ and the straight lines $x = 1$, $x = 4$ and $y = 0$, **find :**

(1) The area of region in square units to the nearest unit.

(2) The volume of the solid generated by revolving the region a about x-axis.

« $4 \ln 4 + \frac{15}{2}$ square unit , 57π cubic unit »

29 📖 Find the volume of the solid generated by revolving the region bounded by the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and x-axis where a and b are two constants a complete revolution about x-axis.

« $\frac{4}{3}\pi b^2 a$ cubic unit »

30 📖 Use the integration to prove that :

(1) Volume of sphere = $\frac{4}{3}\pi r^3$ (r is the radius length of the sphere).

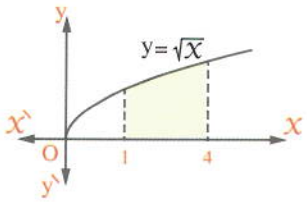
(2) Volume of the right circular cylinder = $\pi r^2 h$ (r is the radius length of the cylinder base and h is its height).

(3) Volume of the right circular cone = $\frac{1}{3}\pi r^2 h$ (r is the radius length of its base and h is its height)

31 Find the volume of the solid generated by revolving the region bounded by the curve : $y = \sqrt{4 - x^2}$ and the straight line $y = 0$ a complete revolution about x-axis.

« $\frac{32}{3}\pi$ cubic unit »

32 **Choose the correct answer from the given ones :**



(1) The volume of the solid generated by revolving the shaded region a complete revolution about x-axis = ... cubic unit.

(a) $\frac{14}{3}\pi$ (b) $\frac{15}{2}\pi$ (c) $\frac{15}{2}$ (d) $\frac{14}{3}$

(2) The volume of the solid generated by revolving the region bounded by the curve $f(x) = x^2$ and x-axis and the two straight lines $x = -2$, $x = 2$ one revolution about x-axis equals ..... cubic unit.

(a) $\frac{16\pi}{5}$ (b) $\frac{32\pi}{5}$ (c) $\frac{64\pi}{5}$ (d) 4π

(3) The volume of the solid generated by revolving the region bounded by the curve $y = x^2$ and the x-axis and the straight line $x = a$ where $a \in \mathbb{R}^+$ a complete revolution about x-axis equals .....

(a) $\frac{1}{2}\pi a^4$ (b) πa^2 (c) $\frac{1}{5}\pi a^5$ (d) $\frac{1}{3}\pi a^3$

(4) The volume of the solid generated by revolving the region bounded by the curve $y^2 = 2ax$ and the straight line $x = b$ (where $a , b , \in \mathbb{R}^*$) , $y = 0$ a half revolution about x-axis equals ..... cubic unit.

(a) $\pi a b^2$ (b) $\pi b a^2$ (c) $\pi a b$ (d) $\pi a^2 b^2$

(5) The volume of the solid generated by revolving the region bounded by the curve $y = \sec x$ and the two straight lines $x = 0$, $x = \frac{\pi}{3}$ a complete revolution about x-axis equals ..... cubic unit.

(a) $\sqrt{3}$ (b) $\frac{1}{\sqrt{3}}$ (c) $\frac{\pi}{\sqrt{3}}$ (d) $\sqrt{3}\pi$

(6) $\pi \int_{-2}^{2} (4 - x^2)\, dx$ is the volume of .....

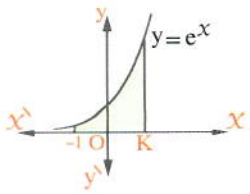
(a) a sphere whose radius length is 4 units.

(b) a right circular cone whose height is 4 units and its base radius 3 units.

(c) a sphere whose radius length is 2 units.

(d) a right circular cylinder whose height is 4 units and its base radius 2 units

(7) In the opposite figure :



If the volume generated by revolving the shaded region a complete revolution about $\mathcal{X}$-axis equals $\frac{\pi}{2}(e^{10} - e^{-2})$ cubic unit , then k = ...

(a) 5 (b) 10 (c) 20 (d) – 5

(8) The ratio between the volume of the solid generated by revolving the curve of the function $y = f(\mathcal{X})$ about $\mathcal{X}$-axis a complete revolution : the volume of the solid generated by revolving the same curve about $\mathcal{X}$-axis two complete revolutions = ...

(a) 1 : 1 (b) 1 : 2 (c) 2 : 1 (d) 1 : 4

(9) The ratio between the volume of the solid generated by revolving the curve of the function $y = f(\mathcal{X})$ about $\mathcal{X}$-axis half revolution : the volume of the solid generated by revolving the same curve about $\mathcal{X}$-axis two and half revolutions = ...

(a) 1 : 1 (b) 1 : 2 (c) 5 : 1 (d) 1 : 5

(10) The volume of the solid generated by revolving circle with equation $(\mathcal{X} - 5)^2 + y^2 = 9$ a complete revolution about $\mathcal{X}$-axis = ... cubic unit.

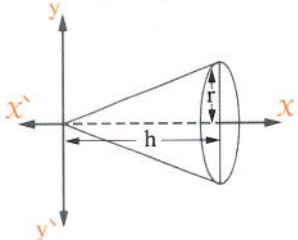
(a) 18π (b) 27π (c) 36π (d) 72π

(11) If A_1 is the area bounded by the curve $y_1 = \sqrt{\mathcal{X}}$, the $\mathcal{X}$-axis and straight line $\mathcal{X} = 2$ and A_2 is the area bounded by the curve $y_2^2 = \mathcal{X}$ and the straight line $\mathcal{X} = 2$, V_1 and V_2 are the volumes of two solids generated by revolving the two regions A_1 , A_2 a complete revolution about $\mathcal{X}$-axis respectively , then .....

(a) $A_1 = A_2$, $V_1 = V_2$ (b) $A_1 = \frac{1}{2} A_2$, $V_1 = \frac{1}{2} V_2$

(c) $A_1 = A_2$, $V_1 = \frac{1}{2} V_2$ (d) $A_1 = \frac{1}{2} A_2$, $V_1 = V_2$

(12) In the opposite figure :



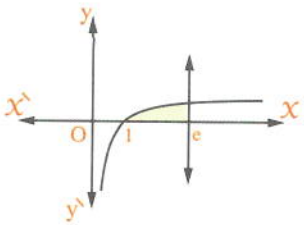
The axis of a right cone lies along the $\mathcal{X}$-axis and its vertex at the origin , then its volume = ...

(a) $\pi \int_0^h \left(\frac{r}{h}\mathcal{X}\right) d\mathcal{X}$ (b) $\pi \int_0^r \left(\frac{h}{r} y\right) dy$

(c) $\pi \int_0^h \left(\frac{r^2}{h^2}\mathcal{X}^2\right) d\mathcal{X}$ (d) $\pi \int_0^r \left(\frac{h^2}{r^2} y^2\right) dy$

(13) The opposite figure represents the curve :

$y = \frac{\ln x}{\sqrt{x}}$ and the line $x = e$, then the volume of the generated solid by revolving the shaded region a complete revolution about the x-axis = ... cube units.



(a) $\frac{1}{3}\pi$ (b) π

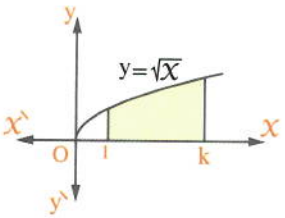
(c) 3π (d) 9π

(14) *(1st Session 2025)* If the volume of the solid generated by revolving the region bounded by the curve $y = \sqrt{6}\, e^{ax}$, and the straight lines $y = 0$, $x = 0$ and $x = 1$, a complete revolution about x-axis equals $\frac{3\pi}{a}(e^6 - 1)$ cubic unit , then a^2 = ... where $a > 0$

(a) 36 (b) 9 (c) 6 (d) 3

33 **In the opposite figure :**

If the volume of the solid generated by revolving the shaded region a complete revolution about x-axis = (12π) cubic unit. Find the value of k.



« 5 »

Higher skills

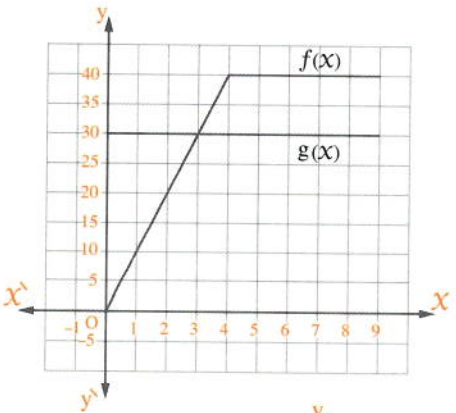
34 **Choose the correct answer from those given :**

(1) The opposite figure represents the curves of two functions f , g in the interval $[0 , 9]$

If $\int_0^a f(x)\, dx = \int_0^a g(x)\, dx$, then a = ...

(a) 3 (b) 4

(c) 5 (d) 8



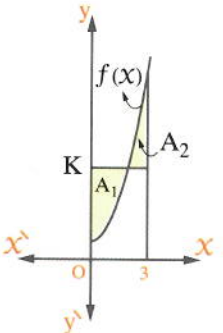
(2) **In the opposite figure :**

$f(x) = x^2 + 1$

, then k which makes $A_1 = A_2$ equals .....

(a) 3 (b) 4

(c) 5 (d) 8

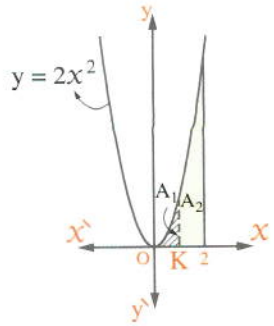


(3) In the opposite figure :

If $A_2 = 7A_1$, then k = ...

(a) $\frac{1}{5}$ (b) $\frac{1}{4}$

(c) 1 (d) $\frac{3}{2}$

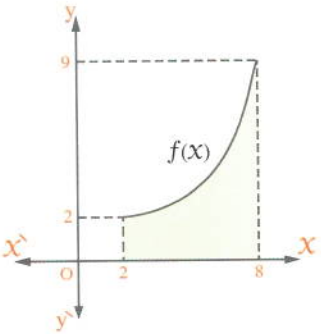


(4) In the opposite figure :

If the curve of the function $f(X)$ is continuous and convex downward in the interval [2 , 8] , then $\int_2^8 f(X)\, dX$ can not be .....

(a) 28 (b) 30

(c) 32 (d) 34



(5) The area of the region bounded by the curve whose parametric equations $y = 3t^2$, $X = 6t$ and the X-axis and the two straight lines $X = 0$, $X = 12$ equals ..... square units.

(a) 48 (b) 96 (c) 132 (d) 192

(6) In the opposite figure :

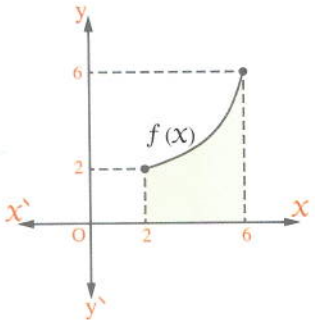
f is a function defined on the interval [2 , 6] and $f(X)$ is a continuous and one-to-one

If the shaded area = 13 square units.

, then $\int_2^6 f^{-1}(X)\, dX$ = ...

(a) 18 (b) 19

(c) 20 (d) 21



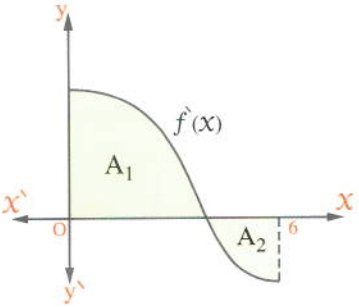
(7) The opposite figure represent the curve of the function $\dot{f}(X)$ in the interval [0 , 6]

If A_1 and A_2 are the shaded regions

$A_1 = 20$ square units.

, $A_2 = 7$ square units. , $f(0) = 2$, then the greatest value of the function f in the interval [0 , 6] equals .....

(a) 13 (b) 15 (c) 22 (d) 29



(8) If $x \in [1 , 4]$ and $f(x) \in [5 , 8]$, then $\int_1^4 f(x)\, dx \in$ .....

(a) $[1 , 4]$ (b) $[5 , 8]$ (c) $[6 , 12]$ (d) $[15 , 24]$

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35 Find the area of the plane region bounded by the two curves :

$y = \sqrt{x}$, $y = \sqrt{4 - 3x}$ and the straight line $y = 0$ « $\frac{8}{9}$ square units »

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36 Find the volume of the solid generated by revolving the region bounded by the curve $y = \frac{1}{x - 1}$ and the two straight line $x = 2$ and $x = 4$ and x-axis a complete revolution about x-axis

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37 ABCD is a quadrilateral in which A $(0 , 0)$, B $(7 , 0)$, C $(2 , 5)$, D $(0 , 3)$ By using integration find the volume of the solid generated by revolving the quadrilateral ABCD a complete revolution about x-axis , also the volume of the solid generated by revolving the triangle BCD a complete revolution about x-axis. « $\frac{223}{3}\pi$, $\frac{160}{3}\pi$ cubic unit »



Notes



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Question Bank & Model Examinations

- **Pure Mathematics**
  (Algebra & Analytic Solid Geometry
  + Differential & Integral Calculus)
- **Applied Mathematics**
  (Statics + Dynamics)

3rd Sec.